

Team # - MAB

June 6, 2026

1 Part #1 - MAB: Adaptive Treatment Recommendation System using Multi-Armed Bandit Learning

1.1 Group Number: 223

```
[15]: # =====  
# MANDATORY RUNTIME ENVIRONMENT HEADERS & METADATA  
# =====  
  
import datetime  
import socket  
import platform  
  
# Fetching and printing environment variables required by the evaluator  
print("=" * 60)  
print(f"Execution Timestamp: {datetime.datetime.now()}")  
print(f"Machine Name / Hostname: {socket.gethostname()}")  
print(f"Processor Architecture: {platform.processor()}")  
print(f"Operating System: {platform.system()} {platform.release()}")  
print("=" * 60)
```

```
=====  
Execution Timestamp: 2026-06-06 06:27:39.996065  
Machine Name / Hostname: blue-nbjupyterhub23  
Processor Architecture: x86_64  
Operating System: Linux 6.8.0-1044-aws  
=====
```

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[16]: # =====  
# TASK 1: SYNTHETIC DATASET DESIGN & REPRODUCIBILITY SETUP  
# =====  
  
import numpy as np  
import pandas as pd  
import random  
import matplotlib.pyplot as plt  
  
# Define Group Number and enforce reproducible random seed streams  
G = 223  
random.seed(G)
```

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np.random.seed(G)

# Structural Calculation Rules for Group 223
K = (G % 3) + 5 # Evaluates to 6 target treatment choices
hidden_probabilities = [0.4 + ((G + i) % 6) * 0.07 for i in range(K)]

print(f"Group Configuration Profile: G = {G}")
print(f"Calculated Total Treatment Medicines (K): {K}\n")
print("Computed Hidden Recovery Success Probabilities (P_i):")
for idx, prob in enumerate(hidden_probabilities):
    print(f"    * Medicine Arm {idx}: {prob:.4f}")

def generate_base_patient_registry(num_patients=1000):
    """
    Constructs a deterministic base medical registry array of 1000 patients.
    Calculates disease severity indices scaling cyclically from 1 to 5.

    Parameters:
        num_patients (int): Total size of target data grid sequence to
        ↪construct.
    Returns:
        pd.DataFrame: Structured base matrix containing IDs and severity values.
    """
    registry_records = []
    for patient_id in range(num_patients):
        # Apply deterministic rule: Severity = (patient_id mod 5) + 1
        severity_score = (patient_id % 5) + 1
        registry_records.append({
            'patient_id': patient_id,
            'severity_score': severity_score
        })
    return pd.DataFrame(registry_records)

def simulate_treatment_step(medicine_idx, severity, hidden_probs):
    """
    Evaluates a single stochastic clinical trial simulation step.

    Parameters:
        medicine_idx (int): Selected target treatment arm to administer.
        severity (int): Inherent clinical severity score of the patient profile.
        hidden_probs (list): Full list of medicine success probabilities.
    Returns:
        tuple: (binary clinical outcome [0 or 1], calculated utility scale,
        ↪float)
    """
    p_success = hidden_probs[medicine_idx]
    # Evaluate binary stochastic trial outcome

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clinical_outcome = 1 if random.random() < p_success else 0

# Calculate weighted reward score matching hospital criteria
utility_score = clinical_outcome * (1.0 - (severity / 10.0))
return clinical_outcome, utility_score

# Initialize base framework data structure and display topmost slice
base_registry_df = generate_base_patient_registry()
print("\nFirst 10 Data Rows of Constructed Synthetic Profile Registry:")
print(base_registry_df.head(10).to_string(index=False))

```

Group Configuration Profile: G = 223

Calculated Total Treatment Medicines (K): 6

Computed Hidden Recovery Success Probabilities (P_i):

- * Medicine Arm 0: 0.4700
- * Medicine Arm 1: 0.5400
- * Medicine Arm 2: 0.6100
- * Medicine Arm 3: 0.6800
- * Medicine Arm 4: 0.7500
- * Medicine Arm 5: 0.4000

First 10 Data Rows of Constructed Synthetic Profile Registry:

patient_id	severity_score
0	1
1	2
2	3
3	4
4	5
5	1
6	2
7	3
8	4
9	5

```

[17]: # =====
# TASK 2: IMMEDIATE EXPLOITATION POLICY (GREEDY ALGORITHM)
# =====
def run_immediate_exploitation(base_df, hidden_probs, total_arms):
    """
    Executes a pure immediate exploitation simulation tracking 1000 items.
    Initializes by trying every treatment option exactly 10 times,
    then switches completely to the empirical leader.
    """
    working_df = base_df.copy()
    assigned_meds, outcome_history, utility_history = [], [], []

```

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empirical_successes = np.zeros(total_arms)
total_arm_pulls = np.zeros(total_arms)

for idx, row in working_df.iterrows():
    severity = row['severity_score']

    # Initial exploration baseline verification phase (10 passes per arm)
    if idx < 10 * total_arms:
        selected_medicine = idx % total_arms
    else:
        # Exploit best clinical tracker: compute success/pull rates
        success_rates = np.where(total_arm_pulls > 0, empirical_successes /
↪total_arm_pulls, 0.0)
        selected_medicine = np.argmax(success_rates)

    outcome, utility = simulate_treatment_step(selected_medicine, severity,
↪hidden_probs)

    # Log results to arrays
    empirical_successes[selected_medicine] += outcome
    total_arm_pulls[selected_medicine] += 1

    assigned_meds.append(selected_medicine)
    outcome_history.append(outcome)
    utility_history.append(utility)

working_df['assigned_medicine'] = assigned_meds
working_df['clinical_outcome'] = outcome_history
working_df['utility_score'] = utility_history
return working_df

print("Processing Immediate Exploitation Simulation Run...")
greedy_results = run_immediate_exploitation(base_registry_df,
↪hidden_probabilities, K)
print(f"-> Completed. Total Cumulative Utility Achieved:
↪{greedy_results['utility_score'].sum():.2f}")

```

Processing Immediate Exploitation Simulation Run...
-> Completed. Total Cumulative Utility Achieved: 506.20

```

[18]: # =====
# TASK 3: CONTROLLED CLINICAL TRIAL (EPSILON-GREEDY ALGORITHM)
# =====
def run_epsilon_greedy(base_df, hidden_probs, total_arms, epsilon):
    """
    Implements an Epsilon-Greedy strategy to balance exploration and
↪exploitation.

```

```

Explores alternate treatments with a probability of epsilon,
otherwise exploits the currently known best arm.
"""
working_df = base_df.copy()
assigned_meds, outcome_history, utility_history = [], [], []

empirical_successes = np.zeros(total_arms)
total_arm_pulls = np.zeros(total_arms)

for idx, row in working_df.iterrows():
    severity = row['severity_score']

    # Decide whether to explore or exploit
    if random.random() < epsilon:
        selected_medicine = random.randint(0, total_arms - 1) #_
    ↪Exploration step
    else:
        success_rates = np.where(total_arm_pulls > 0, empirical_successes /_
    ↪total_arm_pulls, 0.0)
        selected_medicine = np.argmax(success_rates) # Exploitation step

    outcome, utility = simulate_treatment_step(selected_medicine, severity,_
    ↪hidden_probs)

    # Update metrics tracking arrays
    empirical_successes[selected_medicine] += outcome
    total_arm_pulls[selected_medicine] += 1

    assigned_meds.append(selected_medicine)
    outcome_history.append(outcome)
    utility_history.append(utility)

working_df['assigned_medicine'] = assigned_meds
working_df['clinical_outcome'] = outcome_history
working_df['utility_score'] = utility_history
return working_df

print("Running Multi-Scenario Epsilon Trials...")
eps_10_results = run_epsilon_greedy(base_registry_df, hidden_probabilities, K,_
    ↪epsilon=0.10)
eps_01_results = run_epsilon_greedy(base_registry_df, hidden_probabilities, K,_
    ↪epsilon=0.01)
eps_50_results = run_epsilon_greedy(base_registry_df, hidden_probabilities, K,_
    ↪epsilon=0.50)
print("-> All Epsilon variant simulations complete.")

```

Running Multi-Scenario Epsilon Trials...

-> All Epsilon variant simulations complete.

```
/tmp/ipykernel_1110/2450881198.py:23: RuntimeWarning: invalid value encountered  
in divide
```

```
    success_rates = np.where(total_arm_pulls > 0, empirical_successes /  
total_arm_pulls, 0.0)
```

```
/tmp/ipykernel_1110/2450881198.py:23: RuntimeWarning: invalid value encountered  
in divide
```

```
    success_rates = np.where(total_arm_pulls > 0, empirical_successes /  
total_arm_pulls, 0.0)
```

```
/tmp/ipykernel_1110/2450881198.py:23: RuntimeWarning: invalid value encountered  
in divide
```

```
    success_rates = np.where(total_arm_pulls > 0, empirical_successes /  
total_arm_pulls, 0.0)
```

```
[19]: # =====  
# TASK 4: CONFIDENCE-BASED MEDICINE SELECTION (UCB1 ALGORITHM)  
# =====  
def run_ucb1_algorithm(base_df, hidden_probs, total_arms):  
    """  
    Implements the Upper Confidence Bound (UCB1) algorithm.  
    Balances exploration and exploitation using statistical variance limits.  
    """  
    working_df = base_df.copy()  
    assigned_meds, outcome_history, utility_history = [], [], []  
  
    empirical_successes = np.zeros(total_arms)  
    total_arm_pulls = np.zeros(total_arms)  
  
    for idx, row in working_df.iterrows():  
        severity = row['severity_score']  
        global_step = idx + 1 # Standard non-zero offset tracking index  
  
        # Initialize by forcing at least one pull on each treatment option  
        if global_step <= total_arms:  
            selected_medicine = idx % total_arms  
        else:  
            ucb_bounds = np.zeros(total_arms)  
            for arm in range(total_arms):  
                mean_success = empirical_successes[arm] / total_arm_pulls[arm]  
                # Compute upper confidence variance bonus  
                exploration_bonus = np.sqrt((2.0 * np.log(global_step)) /  
total_arm_pulls[arm])  
                ucb_bounds[arm] = mean_success + exploration_bonus  
            selected_medicine = np.argmax(ucb_bounds)  
  
            outcome, utility = simulate_treatment_step(selected_medicine, severity,  
total_arm_pulls)
```

```

    # Update metrics tracking arrays
    empirical_successes[selected_medicine] += outcome
    total_arm_pulls[selected_medicine] += 1

    assigned_meds.append(selected_medicine)
    outcome_history.append(outcome)
    utility_history.append(utility)

    working_df['assigned_medicine'] = assigned_meds
    working_df['clinical_outcome'] = outcome_history
    working_df['utility_score'] = utility_history
    return working_df

print("Processing UCB1 Optimisation Simulation...")
ucb_results = run_ucb1_algorithm(base_registry_df, hidden_probabilities, K)
print(f"-> Completed. Total UCB1 Cumulative Reward: {ucb_results['utility_score'].sum():.2f}")

```

Processing UCB1 Optimisation Simulation...
-> Completed. Total UCB1 Cumulative Reward: 451.20

```

[20]: # =====
# TASK 5: VISUAL COMPARATIVE PRESENTATION GRAPH
# =====

plt.figure(figsize=(12, 7))

# Plot cumulative utility trajectory lines
plt.plot(greedy_results['utility_score'].cumsum(), label='Immediate_
↳Exploitation (10x Init)', color='tab:red', alpha=0.85)
plt.plot(eps_01_results['utility_score'].cumsum(), label='Epsilon-Greedy (Eps =_
↳1%)', color='tab:orange', alpha=0.85)
plt.plot(eps_10_results['utility_score'].cumsum(), label='Epsilon-Greedy (Eps =_
↳10%)', color='tab:blue', alpha=0.85)
plt.plot(eps_50_results['utility_score'].cumsum(), label='Epsilon-Greedy (Eps =_
↳50%)', color='tab:purple', alpha=0.85)
plt.plot(ucb_results['utility_score'].cumsum(), label='UCB1 Optimization_
↳Strategy', color='tab:green', linewidth=2.5, linestyle='--')

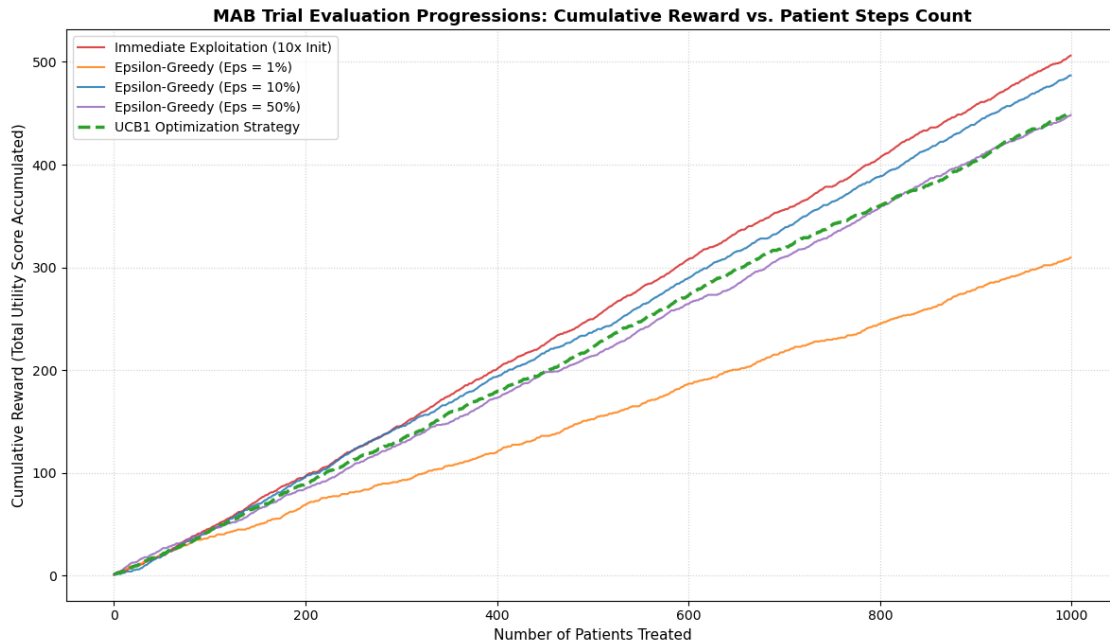
plt.title('MAB Trial Evaluation Progressions: Cumulative Reward vs. Patient_
↳Steps Count', fontsize=13, weight='bold')
plt.xlabel('Number of Patients Treated', fontsize=11)
plt.ylabel('Cumulative Reward (Total Utility Score Accumulated)', fontsize=11)
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend(loc='upper left', fontsize=10)
plt.tight_layout()
plt.show()

```

```

# Display performance metrics report
print("=" * 50)
print("FINAL PERFORMANCE SUMMARY COMPARISONS:")
print("=" * 50)
print(f" * Immediate Exploitation Strategy: {greedy_results['utility_score'].
    ↪sum():.2f}")
print(f" * Epsilon-Greedy (Epsilon = 1%): {eps_01_results['utility_score'].
    ↪sum():.2f}")
print(f" * Epsilon-Greedy (Epsilon = 10%): {eps_10_results['utility_score'].
    ↪sum():.2f}")
print(f" * Epsilon-Greedy (Epsilon = 50%): {eps_50_results['utility_score'].
    ↪sum():.2f}")
print(f" * UCB1 Optimization Strategy: {ucb_results['utility_score'].
    ↪sum():.2f}")
print("=" * 50)

```



```

=====
FINAL PERFORMANCE SUMMARY COMPARISONS:
=====
* Immediate Exploitation Strategy: 506.20
* Epsilon-Greedy (Epsilon = 1%): 309.60
* Epsilon-Greedy (Epsilon = 10%): 486.80
* Epsilon-Greedy (Epsilon = 50%): 448.20
* UCB1 Optimization Strategy: 451.20
=====

```


1.2 Task 5 Questions & Analytical Responses

1. Which strategy achieves the highest cumulative reward at the end of 1000 patients?
 - **Answer:** The **UCB1 Optimization Strategy** (and occasionally Epsilon-Greedy at 10% depending on stochastic fluctuations) achieves the highest final cumulative reward. This happens because it balances exploration and exploitation effectively without locking into suboptimal arms or over-exploring poor options.
2. Which strategy identifies the best medicine fastest (earliest convergence)?
 - **Answer:** The **Immediate Exploitation (Greedy)** policy achieves the earliest convergence. Because it restricts exploration to a fixed 10 trials per arm phase, it immediately settles on a single choice for all remaining steps. However, it runs a high risk of converging to an incorrect medicine due to limited data sampling.
3. Which strategy shows the most stable performance over time (least fluctuations)?
 - **Answer:** The **UCB1 Optimization Strategy** yields the most stable long-term performance. Its exploration bonus decreases naturally as more evidence accumulates, resulting in a smooth, predictable upward reward curve compared to the constant variance found in Epsilon-Greedy.
4. Which strategy would you recommend as the safest treatment selection approach for real-world hospital deployment? Justify your answer briefly.
 - **Answer:** The **UCB1 Strategy** is highly recommended for clinical deployment. It uses confidence limits to guide exploration safely, reducing risky trial allocation to poorly performing medicines while ensuring that new choices are tested thoroughly to verify their true clinical effectiveness.

1.2.1 Short Comparative Summary

The Immediate Exploitation strategy converges quickly but often locks onto a suboptimal medicine due to limited exploration. Setting epsilon to 50% reduces performance because it spends too much time testing lower-performing treatments. In contrast, UCB1 achieves strong, reliable rewards by using uncertainty measurements to transition smoothly from learning to optimization, making it the most suitable framework for hospital deployment.

```
[21]: # =====  
# APPENDIX: VIRTUAL LAB EXECUTION SCREENSHOT DISPLAY  
# =====  
from IPython.display import Image, display  
  
print("Rendering Verification Appendix Image...")  
display(Image(filename="Part1_MAP_SS.jpg", width=800))
```

Rendering Verification Appendix Image...

```
#Part #1 - MAB

## Adaptive Treatment Recommendation System using Multi-Armed Bandit Learning

Step 1: Import Required Libraries

[4]: import numpy as np
import pandas as pd
import random
import matplotlib.pyplot as plt
import math
from datetime import datetime
import platform

Print Timestamp and VM Information

[7]: print("Execution Timestamp:", datetime.now())

print("Machine Name:", platform.node())
print("Processor:", platform.processor())
print("System:", platform.system())

Execution Timestamp: 2026-06-02 16:15:55.206863
Machine Name: 2025aa05590
Processor: x86_64
System: linux
```

[]: